

IN-HOUSE SOLDER PASTE & FLUX

Complete Technical Guide

SAC305 • ALPHA WS-820 • ALPHA OM-353

ALPHA RF-800 • AIM WS715M • SN100C

ALPHA WS-809 • ALPHA OM-5100

PCBA In-House Materials Inventory & Process Standards

RoHS Compliant | Non-RoHS | Flux Systems | Process Compatibility

1. IN-HOUSE SOLDER PASTE & FLUX OVERVIEW

This guide documents all solder pastes and fluxes currently stocked and used in-house for PCBA manufacturing operations. Materials are categorized by RoHS compliance status, alloy composition, flux chemistry, and intended process application. Proper material selection and process control ensure compliance with customer requirements (Tesla, Zoox, Probe services) and industry standards (IPC/J-STD-001, IPC-A-610, RoHS Directive 2011/65/EU).

In-House Material Categories:

- **RoHS Lead-Free Solder Pastes:** SAC305 alloys with no-clean and water-soluble fluxes
- **Non-RoHS Leaded Solder Pastes:** SnPb alloys for exempt applications and legacy repair
- **Liquid Fluxes:** No-clean and water-soluble for wave soldering and selective soldering
- **Bar Solders:** SAC305 and SN100C for wave solder pot maintenance

■ **Key Principle:** Leaded and lead-free materials must be physically separated at all times. Cross-contamination of even trace lead into a RoHS process voids compliance and can cause catastrophic joint failures. Color-coded storage (GREEN = lead-free, RED = leaded) is mandatory.

2. SAC305 LEAD-FREE ALLOY

2.1 SAC305 Alloy Composition

SAC305 is the primary lead-free alloy used in-house. It is a tin-silver-copper alloy that meets all major environmental and regulatory requirements for modern electronics manufacturing.

Composition:

- **Tin (Sn):** 96.5%
- **Silver (Ag):** 3.0%
- **Copper (Cu):** 0.5%

Regulatory Compliance:

- **RoHS:** Compliant — Lead content < 0.1% (1000 ppm)
- **REACH:** Compliant — No SVHC substances above threshold
- **JEIDA:** Compliant — Japanese Electronics Industry standards
- **IPC/J-STD-006:** Meets requirements for electronic grade solder alloys

Physical Properties:

- **Melting Point:** 217–220°C (eutectic range)
- **Recommended Reflow Peak:** 245–250°C
- **Time Above Liquidus (TAL):** 30–90 seconds
- **Joint Appearance:** Matte, grainy (normal for lead-free)
- **Density:** ~7.4 g/cm³

Flux Compatibility:

SAC305 bar solder is compatible with all major brands of **no-clean** and **water-soluble** electronic grade fluxes. In-house, we pair SAC305 with:

- **ALPHA WS-820** — Water-soluble, halide-free (primary RoHS paste)
- **ALPHA OM-353** — No-clean, low residue (secondary RoHS paste)
- **AIM WS715M** — Water-soluble liquid flux (wave soldering)
- **ALPHA RF-800** — No-clean liquid flux (wave/selective soldering)

■ **Process Note:** SAC305 exhibits higher copper dissolution rates (2.1× vs SnPb) during wave soldering. Monitor copper content in wave solder pots weekly. Add pure tin or fresh alloy to maintain composition within specification.

3. ALPHA WS-820 LEAD-FREE SOLDER PASTE

3.1 Product Identification

ALPHA® WS-820 is the newest Alpha brand lead-free, halide-free solder paste in the in-house inventory. It is the **primary lead-free water-soluble paste** used for RoHS-compliant SMT reflow operations.

Product Code: ALPHAWS 820 Solder Paste (LF-WS)

- **Alloy:** SAC305 (96.5Sn / 3.0Ag / 0.5Cu)
- **Flux Type:** Water-soluble, halide-free
- **IPC Classification:** ORH0 / ORH1 (Organic, halide-free)
- **RoHS Status:** Fully compliant

Key Features:

- **Ideal printability:** Excellent stencil release and response-to-pause
- **Wide reflow profile window:** Forgiving process parameters
- **Exceptional post-reflow cleanability:** Residues wash easily with DI water
- **Low BGA voiding:** Optimized flux chemistry reduces trapped gas
- **Halide-free:** No corrosive halide activators

Applications:

- Primary paste for all RoHS SMT reflow operations
- High-reliability assemblies requiring post-reflow cleaning
- BGA and CSP packages where low voiding is critical
- Automotive and industrial applications (Tesla, Zoox compliance)

Process Parameters:

- **Storage:** 0–10°C refrigerated, sealed
- **Shelf Life:** 6 months (refrigerated)
- **Thaw Time:** 4 hours minimum at room temperature
- **Stencil Life:** 4–8 hours at 20–25°C
- **Reflow Peak:** 245–250°C for SAC305 alloy
- **Cleaning:** DI water wash at 60–70°C recommended

■ **Best Practice:** WS-820's excellent cleanability makes it ideal for assemblies with high-impedance circuits or under-fill requirements where flux residue must be completely removed.

4. ALPHA OM-353 SAC305 NO-CLEAN SOLDER PASTE

4.1 Product Identification

ALPHA® OM-353 is the **secondary lead-free no-clean solder paste** in the in-house inventory. It is used when water washing is not desired or when benign residue can remain on the board without reliability concerns.

Product Specifications:

- **Alloy:** SAC305 (96.5Sn / 3.0Ag / 0.5Cu)
- **Flux Type:** No-clean, low residue
- **IPC Classification:** REL0 / REL1 (Rosin/Resin, low residue)
- **RoHS Status:** Fully compliant

Key Features:

- **Low residue:** Minimal post-reflow residue for no-clean process
- **Good wetting:** Reliable performance on OSP and ENIG surfaces
- **Wide process window:** Tolerant of profile variations
- **Compatible with:** Standard no-clean reflow profiles

Applications:

- No-clean SMT assemblies where washing is eliminated
- Consumer electronics with cost-sensitive processes
- Assemblies with conformal coating (residue must be compatible)
- High-volume production where throughput is priority

Process Parameters:

- **Storage:** 0–10°C refrigerated, sealed
- **Shelf Life:** 6 months (refrigerated)
- **Thaw Time:** 4 hours minimum at room temperature
- **Stencil Life:** 4–8 hours at 20–25°C
- **Reflow Peak:** 245–250°C for SAC305 alloy
- **Cleaning:** Not required — no-clean process

■ **Caution:** No-clean residues can interfere with conformal coating adhesion and electrical testing (bed-of-nails ICT). Verify compatibility with downstream processes before specifying OM-353.

5. NON-ROHS (LEADED) SOLDER PASTES

These materials are reserved for exempt applications (medical, military, aerospace, legacy repair) and must be physically separated from RoHS materials at all times.

5.1 ALPHA WS-809 — Tin-Lead Water-Soluble

ALPHA® WS-809 is the in-house **non-RoHS water-soluble solder paste**. It uses traditional SnPb alloy with a water-washable flux system for exempt applications requiring complete residue removal.

Product Specifications:

- **Alloy:** Sn63Pb37 or Sn62Pb36Ag2 (verify jar label)
- **Flux Type:** Water-soluble, halide-containing (M1 classification)
- **IPC Classification:** ORM1 / WSM1
- **RoHS Status:** NON-COMPLIANT — exempt use only

Applications:

- Medical device assemblies (RoHS exempt category)
- Military and aerospace electronics (MIL-STD compliance)
- Repair and rework of pre-2006 equipment
- Any application where SnPb is contractually specified

Process Parameters:

- **Storage:** 0–10°C refrigerated
- **Shelf Life:** 6 months
- **Reflow Peak:** 210–230°C
- **Cleaning:** DI water wash mandatory — residues are corrosive

5.2 ALPHA OM-5100 — Tin-Lead No-Clean

ALPHA® OM-5100 is the in-house **non-RoHS no-clean solder paste**. It provides traditional SnPb soldering with minimal residue for exempt applications where washing is not required.

Product Specifications:

- **Alloy:** Sn63Pb37 or Sn60Pb40 (verify jar label)
- **Flux Type:** No-clean, rosin-based (RMA)
- **IPC Classification:** ROL0 / ROL1
- **RoHS Status:** NON-COMPLIANT — exempt use only

Applications:

- Legacy product manufacturing (contractual SnPb requirement)
- Prototype and low-volume exempt assemblies
- Repair operations where no-clean is preferred

Process Parameters:

- **Storage:** 0–10°C refrigerated
- **Shelf Life:** 6 months
- **Reflow Peak:** 210–230°C
- **Cleaning:** Not required — no-clean process

■■ **CRITICAL SEPARATION:** WS-809 and OM-5100 must be stored in RED-labeled zones only. Never use leaded stencils, squeegees, or tools on lead-free lines. XRF verification required for all exempt production lots.

6. LIQUID FLUXES FOR WAVE & SELECTIVE SOLDERING

6.1 ALPHA RF-800 — No-Clean Liquid Flux

ALPHA® RF-800 is the primary **no-clean liquid flux** used for wave soldering and selective soldering operations. It is compatible with both lead-free (SAC305, SN100C) and leaded (SnPb) bar solders in the wave solder pot.

Product Specifications:

- **Flux Type:** No-clean, alcohol-based
- **IPC Classification:** ROL0, ROL1, REL0, or REL1 (verify batch certificate)
- **Application:** Spray, foam, or brush application
- **Solvent:** Isopropyl alcohol (IPA) based
- **Residue:** Non-corrosive, electrically benign

Key Features:

- **Wide process window:** Tolerant of preheat and wave variations
- **Good wetting:** Reliable on OSP, ENIG, and immersion tin surfaces
- **Low residue:** Minimal post-solder residue for no-clean process
- **Compatible with:** SAC305, SN100C, and SnPb bar solders

Applications:

- Wave soldering with SAC305 or SN100C bar solder
- Selective soldering operations
- No-clean process where post-solder washing is eliminated
- High-volume through-hole assembly

Process Parameters:

- **Storage:** 10–25°C, sealed container, away from ignition sources
- **Shelf Life:** 12 months (unopened)
- **Specific Gravity:** ~0.80–0.85 (verify TDS)
- **Application:** Spray fluxer at 500–1500 mm/min conveyor speed
- **Preheat:** 100–130°C top side before wave

6.2 AIM WS715M — Water-Soluble Liquid Flux

AIM WS715M is the primary **water-soluble liquid flux** used for clean-process wave soldering. It is a neutral pH flux compatible with both lead-free and tin-lead bar solders.

Product Specifications:

- **Flux Type:** Water-soluble, neutral pH
- **IPC Classification:** ORM0 / ORM1 (Organic, moderate activity)

- **Application:** Foam, spray, brush, or dip
- **RoHS Status:** Compatible with lead-free and leaded processes

Key Features:

- **Neutral pH:** Non-corrosive before and after soldering
- **Lead-free and tin-lead compatible:** Universal flux
- **Excellent wetting:** Strong activators for difficult surfaces
- **Wide process window:** Tolerant of preheat variations
- **Low foaming:** Suitable for high-pressure spray applications

Applications:

- Wave soldering with clean process (water wash after soldering)
- High-reliability assemblies requiring zero flux residue
- Automotive and military applications
- Assemblies with conformal coating (residue must be removed)

Process Parameters:

- **Storage:** 10–25°C, sealed container
- **Shelf Life:** 12 months (unopened)
- **Specific Gravity:** ~1.00–1.05 (verify TDS)
- **Application:** Foam fluxer or spray fluxer
- **Cleaning:** DI water wash at 60–70°C mandatory

■ **Selection Guide:** Use RF-800 for no-clean wave soldering (no wash step). Use WS715M for clean-process wave soldering (wash required). Both are compatible with in-house SAC305 and SN100C bar solders.

7. SN100C BAR SOLDER

7.1 Product Identification

SN100C (Part Number: **R1459592X**) is an in-house lead-free bar solder alloy. It is a tin-copper-nickel-germanium alloy developed as a cost-effective alternative to SAC305 for wave soldering operations.

Composition:

- **Tin (Sn):** ~99.0%
- **Copper (Cu):** ~0.7%
- **Nickel (Ni):** ~0.05%
- **Germanium (Ge):** ~0.005%

Regulatory Compliance:

- **RoHS:** Fully compliant — no lead, no cadmium, no mercury
- **REACH:** Compliant
- **JEIDA:** Compliant

Physical Properties:

- **Melting Point:** 227°C
- **Wave Solder Pot:** 255–265°C
- **Joint Appearance:** Bright, shiny (superior to SAC305)
- **Copper Dissolution:** Low — better than SAC305
- **Dross Formation:** Low with proper maintenance
- **Wetting:** Good — improved by nickel and germanium additions

Key Advantages vs SAC305:

- **Lower cost:** No silver content — significantly cheaper than SAC305
- **Better cosmetics:** Shiny joints similar to SnPb appearance
- **Lower copper dissolution:** Extends pot life and reduces maintenance
- **Less dross:** Germanium stabilizes surface oxide
- **No silver scavenging:** Safe for silver-plated components

Applications:

- Wave soldering of through-hole components
- Cost-sensitive high-volume production
- Applications where shiny joint appearance is required
- Assemblies with silver-plated terminations

Process Parameters:

- **Pot Temperature:** 255–265°C
- **Contact Time:** 3–5 seconds
- **Flux Pairing:** RF-800 (no-clean) or WS715M (water-wash)
- **Pot Analysis:** Weekly copper and nickel content checks
- **Dross Skimming:** Daily maintenance required

■ **Recommendation from Sir. Imtiaz:** SN100C is the preferred wave solder alloy for cost-optimized RoHS production. Its shiny appearance simplifies visual inspection, and the lower copper dissolution rate reduces pot maintenance frequency compared to SAC305.

8. IN-HOUSE MATERIAL COMPARISON MATRIX

Material	Type	Alloy/Composition	Flux	RoHS	Primary Use
ALPHA WS-820	Solder Paste	SAC305 96.5Sn/3.0Ag/0.5Cu	WS, Halide-Free	■ Yes	Primary LF SMT Reflow
ALPHA OM-353	Solder Paste	SAC305 96.5Sn/3.0Ag/0.5Cu	NC, Low Residue	■ Yes	Secondary LF SMT Reflow
ALPHA WS-809	Solder Paste	Sn63Pb37 or Sn62Pb36Ag2	WS, M1	■ No	Exempt SMT (Medical/Mil)
ALPHA OM-5100	Solder Paste	Sn63Pb37 or Sn60Pb40	NC, RMA	■ No	Exempt No-Clean SMT
ALPHA RF-800	Liquid Flux	Alcohol-based No-clean	ROL0/ROL1 REL0/REL1	■ Compatible	Wave/Selective Solder
AIM WS715M	Liquid Flux	Water-based Neutral pH	ORM0/ORM1	■ Compatible	Clean Process Wave
SN100C (R1459592X)	Bar Solder	SnCuNiGe ~99Sn/0.7Cu	N/A (Bar)	■ Yes	Wave Solder Pot
SAC305	Bar Solder	96.5Sn/3.0Ag/0.5Cu	N/A (Bar)	■ Yes	Wave Solder Pot (Alt)

Table 1: In-House Solder Paste & Flux Comparison Matrix

9. STORAGE, HANDLING & TRACEABILITY

9.1 Solder Paste Storage

All solder pastes (lead-free and leaded) require refrigerated storage to maintain chemical stability and printing performance.

Lead-Free Pastes (WS-820, OM-353):

- **Location:** GREEN-labeled refrigerator
- **Temperature:** 0–10°C
- **Shelf Life:** 6 months from manufacture date
- **Thaw Time:** 4 hours minimum at 20–25°C
- **Stencil Life:** 4–8 hours after print start

Leaded Pastes (WS-809, OM-5100):

- **Location:** RED-labeled refrigerator (separate from lead-free)
- **Temperature:** 0–10°C
- **Shelf Life:** 6 months from manufacture date
- **Thaw Time:** 4 hours minimum at 20–25°C
- **Stencil Life:** 8–12 hours (longer than lead-free)

9.2 Liquid Flux Storage

- **Location:** Ambient storage cabinet (10–25°C)
- **Shelf Life:** 12 months (unopened)
- **Safety:** RF-800 is alcohol-based — store away from ignition sources
- **Container:** Keep sealed when not in use to prevent solvent evaporation

9.3 Bar Solder Storage

- **Location:** Dry storage area at ambient temperature
- **Shelf Life:** Indefinite (solid metal does not degrade)
- **SAC305 Bars:** GREEN zone
- **SN100C Bars (R1459592X):** GREEN zone
- **SnPb Bars (if any):** RED zone — physically separated

9.4 FIFO & Traceability

- **FIFO:** First-In-First-Out mandatory for all paste jars
- **Labeling:** Every jar marked with receipt date, open date, expiration
- **Batch Records:** Paste lot number recorded per work order
- **XRF Log:** Random joint analysis logged for RoHS compliance verification

■■ **CRITICAL:** Never mix leaded and lead-free solder pastes, fluxes, or tools. Even trace lead contamination (> 1000 ppm) in a RoHS process voids compliance. Implement barcode scanning for paste jars to prevent human error during material selection.

10. PROCESS-MATERIAL PAIRING GUIDE

Correct material pairing ensures optimal solder joint quality, process efficiency, and regulatory compliance. Use this guide to select the right material for each manufacturing process.

10.1 SMT Reflow — RoHS Production

- **Paste:** ALPHA WS-820 (water-soluble) or ALPHA OM-353 (no-clean)
- **Alloy:** SAC305
- **Profile:** Peak 245–250°C, TAL 30–90 sec
- **Cleaning:** DI water wash (WS-820) or no-clean (OM-353)
- **Inspection:** AOI + X-ray for BGA/CSP

10.2 SMT Reflow — Non-RoHS (Exempt)

- **Paste:** ALPHA WS-809 (washable) or ALPHA OM-5100 (no-clean)
- **Alloy:** Sn63Pb37 or Sn62Pb36Ag2
- **Profile:** Peak 210–230°C, TAL 30–60 sec
- **Cleaning:** DI water wash (WS-809) or no-clean (OM-5100)
- **Inspection:** AOI + X-ray; XRF for compliance verification

10.3 Wave Soldering — RoHS

- **Bar Solder:** SN100C (R1459592X) — preferred; SAC305 — alternative
- **Flux:** ALPHA RF-800 (no-clean) or AIM WS715M (water-wash)
- **Pot Temperature:** 255–265°C (SN100C) or 250–260°C (SAC305)
- **Preheat:** 100–130°C
- **Contact Time:** 3–5 seconds
- **Cleaning:** None (RF-800) or DI water wash (WS715M)

10.4 Wave Soldering — Non-RoHS (Exempt)

- **Bar Solder:** Sn63Pb37 or Sn60Pb40
- **Flux:** ALPHA RF-800 (no-clean) or AIM WS715M (water-wash)
- **Pot Temperature:** 240–250°C
- **Preheat:** 80–120°C
- **Contact Time:** 2–4 seconds
- **Cleaning:** None (RF-800) or DI water wash (WS715M)

10.5 Selective Soldering

- **Flux:** ALPHA RF-800 (spray application)
- **Bar Solder:** Match line alloy (SN100C, SAC305, or SnPb)
- **Nozzle Temperature:** 280–320°C (depending on alloy)
- **Dwell Time:** 2–5 seconds per joint

■ **Best Practice:** For Tesla and Zoox automotive production, always use WS-820 or OM-353 with SAC305 alloy. SN100C (R1459592X) is acceptable for wave soldering if customer approves. Never use leaded materials on automotive contracts unless explicitly specified and exempted in writing.

11. QUICK REFERENCE SUMMARY

RoHS Lead-Free SMT Pastes:

- **ALPHA WS-820** — Primary water-soluble SAC305 paste (halide-free, low BGA voiding)
- **ALPHA OM-353** — Secondary no-clean SAC305 paste (low residue)

Non-RoHS Leaded SMT Pastes:

- **ALPHA WS-809** — Water-soluble SnPb paste (exempt applications only)
- **ALPHA OM-5100** — No-clean SnPb paste (exempt applications only)

Liquid Fluxes:

- **ALPHA RF-800** — No-clean flux for wave/selective (ROL0/ROL1/REL0/REL1)
- **AIM WS715M** — Water-soluble flux for clean process (neutral pH, foam/spray/brush)

Bar Solders:

- **SN100C (R1459592X)** — Preferred lead-free wave solder (shiny joints, low cost)
- **SAC305** — Alternative lead-free wave solder (high reliability)

Storage Rules:

- **Pastes:** 0–10°C refrigerated, 6-month shelf life, 4-hour thaw
- **Fluxes:** 10–25°C ambient, 12-month shelf life, sealed
- **Bars:** Dry ambient, indefinite shelf life
- **FIFO:** Mandatory for all paste inventory

Safety & Separation:

- **GREEN zone:** Lead-free materials (WS-820, OM-353, SN100C, SAC305)
- **RED zone:** Leaded materials (WS-809, OM-5100, SnPb bars)
- **Never mix** tools, stencils, or pots between zones
- **XRF audit** random joints for RoHS compliance verification

References & Standards:

- IPC/J-STD-005: Requirements for Soldering Pastes
- IPC/J-STD-006: Requirements for Electronic Grade Solder Alloys
- IPC/J-STD-004: Requirements for Soldering Fluxes
- IPC-A-610: Acceptability of Electronic Assemblies
- RoHS Directive 2011/65/EU
- ALPHA Technical Datasheets: WS-820, OM-353, WS-809, OM-5100, RF-800
- AIM Technical Datasheet: WS715M
- Nihon Superior SN100C Technical Documentation

End of Document — In-House Solder Paste & Flux Complete Guide